Chapter 13 — Halogens

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 294–296

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Which halogen molecule has the strongest bond?

(a) F_2 **(b)** Br_2
(c) I_2 **(d)** Cl_2

Answer: (d) Cl_2 — Cl_2 has the strongest X–X bond; F_2 is anomalously weak due to lone-pair repulsion.

II. The volatility of the halogens generally _____ down the group.

(a) Increases **(b)** Decreases
(c) Remains the same **(d)** Fluctuates unpredictably

Answer: (b) Decreases — Larger molecules have stronger London forces, so volatility decreases.

III. Which halogen molecule has the strongest oxidizing power?

(a) Br_2 **(b)** F_2
(c) I_2 **(d)** Cl_2

Answer: (b) F_2 — F_2 is the strongest oxidizing agent (most easily reduced).

IV. The decreasing thermal stability of the halogens down the group is primarily due to:

(a) Increasing electronegativity of the atoms. **(b)** Decreasing bond length between the halogen atoms.
(c) Increasing atomic radius, leading to a weaker covalent bond. **(d)** Increasing strength of van der Waals forces

Answer: (c) Increasing atomic radius, leading to a weaker covalent bond. — Larger atoms form longer, weaker X–X bonds (F_2 is an exception).

V. Which halide has the strongest reducing power?

(a) F^- **(b)** Cl
(c) Br^- **(d)** I

Answer: (d) I — I^- loses an electron most easily (largest, least strongly held), so it is the strongest reducing agent.

VI. Which statement about the reaction between halogens and hydrogen is correct?

(a) Iodine reacts most vigorously with hydrogen. **(b)** Chlorine and hydrogen explode in darkness.
(c) Fluorine combines explosively with hydrogen even in cold and dark conditions. **(d)** Bromine and hydrogen do not react at all.

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Answer: (c) Fluorine combines explosively with hydrogen even in cold and dark conditions. — F_2 reacts explosively with H_2 even in the cold/dark; reactivity decreases down the group.

VII. How does the acidic strength of hydrogen halides change down the group?

(a) It remains constant.
(b) It decreases from HF to HI.
(c) It increases from HF to HI.
(d) It fluctuates erratically.

Answer: (c) It increases from HF to HI. — H–X bond strength decreases down the group, so HX ionizes more readily: HF < HCl < HBr < HI.

VIII. Why is fluorine the most reactive halogen?

(a) Bond length in the halogen molecule
(b) Bond strength in the halogen molecule
(c) Electronegativity of the halogen
(d) Number of electrons in the halogen molecule

Answer: (c) Electronegativity of the halogen — Fluorine's very high electronegativity and weak F–F bond make it extremely reactive.

IX. A cream precipitate forms when $AgNO_3$ is added to a bromide solution. Its solubility in aqueous ammonia is:

(a) Soluble in dilute ammonia solution
(b) Partially soluble in dilute ammonia solution
(c) Insoluble in dilute ammonia solution
(d) Soluble only upon heating with ammonia

Answer: (c) Insoluble in dilute ammonia solution — AgBr is insoluble in dilute NH_3 but dissolves in concentrated NH_3.

X. Concentrated H_2SO_4 is added to solid NaCl. The initial observation is:

(a) Reddish-brown fumes are evolved.
(b) A purple vapor is evolved.
(c) Steamy white fumes of hydrogen chloride are evolved.
(d) A black solid is formed.

Answer: (c) Steamy white fumes of hydrogen chloride are evolved. — $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$; HCl gives steamy white fumes.

Q.2 — Short Answer Questions

a. Which halogen is the least reactive, which is the most reactive? Give reason.

Most reactive: F_2 (highest electronegativity, weak F–F bond). Least reactive: I_2 (largest size, lowest electronegativity, weakest oxidizing power).

b. For $Br_2 + 2I^- \rightarrow 2Br^- + I_2$, explain which species is oxidized and why.

I^- is oxidized to I_2 because it loses electrons (oxidation number rises from −1 to 0). Br_2 is reduced to Br^-.

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c. What is the role of London dispersion forces in the trend of volatility of halogens?

Down the group, molecules become larger and more polarisable, so London forces strengthen. More energy is needed to separate molecules, so volatility decreases (F_2 gas → I_2 solid).

d. How does the reactivity of halogens with hydrogen vary?

Reactivity decreases down the group: F_2 reacts explosively (cold/dark), Cl_2 in sunlight, Br_2 on heating, I_2 slowly and reversibly with a catalyst.

e. Which halogen is used as an antiseptic? How does it work?

Iodine (as tincture of iodine or povidone-iodine) is used as an antiseptic. It oxidises/iodinates microbial proteins and enzymes, killing microorganisms.

f. What is the colour change when chlorine displaces bromine?

The colourless/pale green chlorine solution turns orange/brown as bromine (Br_2) is formed: $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$.

g. How are the halogen acids ionized in water?

$HX + H_2O \rightarrow H_3O^+ + X^-$. HF ionizes partially (weak acid); HCl, HBr and HI ionize almost completely (strong acids).

h. Why is HF a weaker acid than HCl?

The H–F bond is very strong (short bond, high bond energy) and F^- is strongly solvated, so HF ionizes less readily than HCl.

i. Describe a simple chemical test to distinguish aqueous KBr and KI.

Add $AgNO_3$ solution: KBr gives a cream precipitate (AgBr) soluble in concentrated NH_3; KI gives a yellow precipitate (AgI) insoluble in NH_3. Alternatively, add chlorine water: KBr gives orange Br_2, KI gives brown/violet I_2.

j. Explain the chemical principles behind chlorine as a disinfectant, with equations.

$Cl_2 + H_2O \rightleftharpoons HCl + HOCl$. $HOCl \rightleftharpoons H^+ + OCl^-$. HOCl is a strong oxidising agent that penetrates microbial cells and oxidises essential components, killing bacteria. It is more effective than OCl^-.

k. Describe one significant disadvantage of chlorine in water purification.

Chlorine can react with organic matter to form harmful chlorinated organic by-products (e.g. trihalomethanes), which are potentially carcinogenic.

l. What is a disproportionation reaction? Give an example.

A reaction in which the same element is both oxidized and reduced. Example: $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$ (Cl goes from 0 to −1 and +1).

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m. Chlorine gas reacts differently with NaOH depending on temperature and concentration.

Cold, dilute NaOH gives chloride + hypochlorite; hot, concentrated NaOH gives chloride + chlorate.

n. Write balanced equations for Cl_2 with: (i) cold, dilute NaOH (ii) hot, concentrated NaOH.

(i) $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$.

(ii) $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$.

o. For each reaction in (l)/(n), identify oxidation states of chlorine and explain why both are disproportionation.

In Cl_2, Cl = 0. In cold alkali: NaCl (Cl = −1, reduced) and NaClO (Cl = +1, oxidized). In hot alkali: NaCl (−1, reduced) and $NaClO_3$ (+5, oxidized). In both cases the same element is simultaneously oxidized and reduced → disproportionation.

Descriptive / Long Questions

Q.3 Describe and explain the relative thermal stabilities of the halogen hydrides in terms of bond strength.

Thermal stability decreases from HF to HI: HF > HCl > HBr > HI. Down the group the H–X bond length increases and bond strength decreases, so less energy is needed to break the bond. HF is the most stable (strong H–F bond); HI is the least stable and decomposes readily on heating.

Q.4 Discuss the relative reactivity of the halogen elements as oxidizing agents. Arrange F_2, Cl_2, Br_2, I_2 in increasing order of oxidizing power.

Oxidizing power decreases down the group because atomic size increases and the tendency to gain electrons decreases. Increasing order of oxidizing power: $I_2 < Br_2 < Cl_2 < F_2$. A more reactive halogen displaces a less reactive halide from solution (e.g. $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$).

Q.5 Discuss the reducing power of halide ions with relevant reactions. Also explain the factors affecting it.

Reducing power increases down the group: $F^- < Cl < Br^- < I$. Larger halide ions lose electrons more easily. With concentrated H_2SO_4: Cl^- gives only HCl (no redox); Br^- reduces H_2SO_4 to SO_2 ($2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$); I^- reduces it further to H_2S ($8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$). Factors: ionic size, bond strength and hydration/electron-loss tendency.

Quick Check Solutions (in-chapter practice boxes)

Quick Check 13.1 (book p. 285)

a. Which halogen elements are radioactive?

Astatine (At) is radioactive; tennessine (Ts) is also synthetic and radioactive.

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b. Reason behind the different colours of halogens.

They absorb different wavelengths of visible light as electrons are excited; down the group the absorbed wavelength increases, so the observed colour deepens (F_2 pale yellow → Cl_2 greenish-yellow → Br_2 red-brown → I_2 violet).

c. Why chlorine is more volatile than bromine and iodine.

Cl_2 is smaller with weaker London dispersion forces, so less energy is needed to separate its molecules; it is a gas while Br_2 is a liquid and I_2 a solid.

Quick Check 13.2 (book p. 287)

a. Why the F–F bond is weaker than Cl–Cl although F is the most electronegative.

The F atoms are very small, so the lone pairs on adjacent atoms repel strongly, weakening the F–F bond.

b. Is the reaction between NaCl(aq) and F_2 possible? Give reason and equation.

Thermodynamically yes: F_2 is a stronger oxidizing agent than Cl_2 and can oxidize Cl^-: $F_2 + 2NaCl \rightarrow 2NaF + Cl_2$. However, in aqueous solution F_2 also reacts violently with water, so the simple displacement is complicated.

c. Relationship between oxidizing power of halogens and standard reduction potentials.

The more positive the standard reduction potential ($X_2 + 2e^- \rightarrow 2X^-$), the stronger the oxidizing power: $F_2 > Cl_2 > Br_2 > I_2$.

Quick Check 13.3 (book p. 289)

a. Why $H_2 + F_2$ is explosive but $H_2 + I_2$ is slow and reversible.

The F–F bond is weak and the H–F bond formed is very strong, so the reaction is highly exothermic and explosive. The H–I bond is weak and the reaction is only slightly exothermic/reversible, so it is slow.

b. Which is more exothermic: $H_2 + Cl_2 \rightarrow 2HCl$ or $H_2 + Br_2 \rightarrow 2HBr$?

$H_2 + Cl_2 \rightarrow 2HCl$ is more exothermic because the H–Cl bond is stronger than H–Br, so more energy is released on bond formation.

c. How thermal stability of hydrogen halides relates to bond dissociation energies.

Higher H–X bond dissociation energy means greater thermal stability: HF > HCl > HBr > HI.

d. Why HF is the most thermally stable hydrogen halide.

The H–F bond is the shortest and strongest (highest bond dissociation energy), so it requires the most energy to break.

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Quick Check 13.4 (book p. 290)

a. Why F^- is a weaker reducing agent than Cl^-.

F^- is very small and holds its electrons tightly (high electronegativity), so it is hard to oxidize; Cl^- loses electrons more easily.

b. Cause of different solubilities of silver halides in aqueous ammonia.

Solubility depends on Ksp and formation of the soluble $[Ag(NH_3)_2]^+$ complex. AgCl (higher Ksp) dissolves in dilute NH_3; AgBr needs concentrated NH_3; AgI (very low Ksp) does not dissolve.

c. Equation for KI with Ag^+ followed by aq. NH_3; observation.

$Ag^+ + I \rightarrow AgI\downarrow$ (yellow). Adding aqueous NH_3 does not dissolve AgI, so the yellow precipitate remains.

Quick Check 13.5 (book p. 293)

a. How KI reacts with conc. H_2SO_4; what it indicates about reducing power of iodide.

$8KI + 9H_2SO_4 \rightarrow 4I_2 + H_2S + 8KHSO_4 + 4H_2O$ (or $8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$). Iodide reduces sulfur from +6 to −2, showing I^- is a strong reducing agent.

b. Show that Cl_2 with cold and hot aqueous KOH is disproportionation.

Cold: $Cl_2 + 2KOH \rightarrow KCl + KClO + H_2O$ (Cl: 0 → −1 and +1). Hot: $3Cl_2 + 6KOH \rightarrow 5KCl + KClO_3 + 3H_2O$ (Cl: 0 → −1 and +5). In both, chlorine is simultaneously oxidized and reduced.

c. HI acts as a strong reducing agent; explain with reactions.

$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$; $8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$. HI reduces H_2SO_4 to SO_2/H_2S while being oxidized to I_2.

Quick Check 13.6 (book p. 294)

a. Why HOCl is a more effective disinfectant than OCl^-.

HOCl is neutral and penetrates microbial cell walls more easily, where it oxidises essential cellular components; OCl^- is charged and less able to enter cells.

b. Factors affecting disinfection of bacteria in water.

pH, chlorine dose, contact time, temperature, and the amount of organic matter/turbidity.

c. Primary active species in chlorination of water; equation for formation.

HOCl and OCl^-. Formation: $Cl_2 + H_2O \rightleftharpoons HCl + HOCl$; $HOCl \rightleftharpoons H^+ + OCl^-$.

Textbook checks flagged in this chapter

• p. 296, Q.2(o) refers to “question l)”, but the relevant equations are in part (n); both are answered together.

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